Education City High School

1.4

Polynomial Functions and Rates of Change

Student Notes and Guided Practice

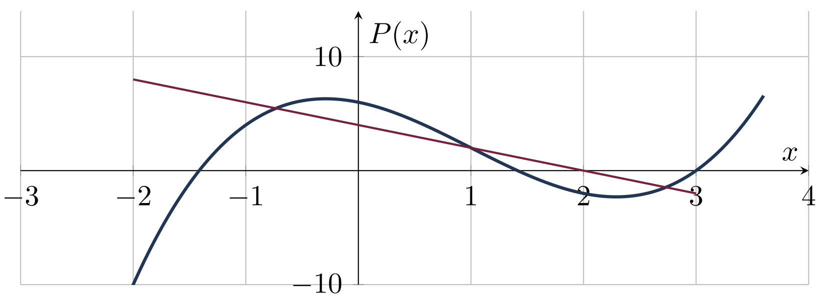
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Analyze polynomial behavior using average rates and finite differences.
- Connect degree to the order of constant finite differences.
- Interpret changing rates in polynomial models.

Essential Question

How does polynomial degree control patterns in successive rates of change?



Concept Framework

Key Idea

For equally spaced inputs, a degree-n polynomial has constant nth differences.

Key Idea

A polynomial rate of change need not be monotonic even when the polynomial is increasing overall.

Key Idea

The leading term dominates global behavior, while all terms influence local rates.

Key Idea

Secant slopes provide interval-based evidence, not instantaneous conclusions.

Formula Map

Formula and Structure

$$\Delta_h P(x) = P(x+h) - P(x)$$

Formula and Structure

$$\deg(\Delta_h P) = \deg(P) - 1 \quad (h \neq 0)$$

Worked Examples

Worked Example 1

Problem. Compute $P(x+1) - P(x)$ for $P(x) = x^3 - 2x$.
Solution. $(x+1)^3 - 2(x+1) - (x^3 - 2x) = 3x^2 + 3x - 1$.

Worked Example 2

Problem. A cubic data table has constant third differences 18 for unit steps. Find the leading coefficient.
Solution. For a cubic, the third difference is $3!a = 6a$. Thus $6a = 18$, so $a = 3$.

Worked Example 3

Problem. Why can two cubic models share end behavior but differ strongly on a measured interval?
Solution. End behavior depends only on degree and leading coefficient. Middle and lower-degree terms can change local shape and rates.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. For $P(x) = x^3 - 2x$, compute $P(x+1) - P(x)$ and state its degree.

2. For $P(x) = 2x^3 + x^2 - 1$, compute $P(x+1) - P(x)$ and state its degree.

3. For $P(x) = -x^4 + 3x^2$, compute $P(x+1) - P(x)$ and state its degree.

4. For $P(x) = 3x^4 - 2x + 5$, compute $P(x+1) - P(x)$ and state its degree.

5. For $P(x) = x^5 - x$, compute $P(x+1) - P(x)$ and state its degree.

6. A quartic table with unit steps has constant fourth difference 96. Find the leading coefficient.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.
2. A restriction I must remember is ____________________.
3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.